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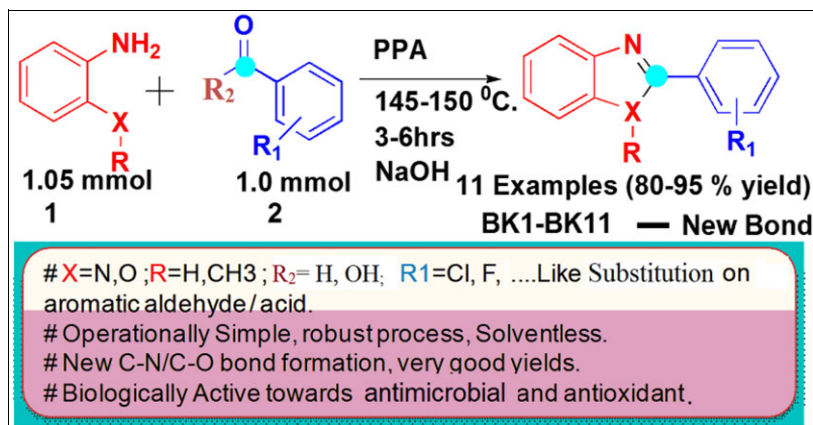
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A series of novel 2-substituted benzimidazole and benzoxazole derivatives as a potential antimicrobial and antioxidant agent were synthesized *via* coupling of *N*-methyl-*o*-phenylenediamine or 2-amino-phenol with aromatic aldehyde and acid in the presence of polyphosphoric acid as an efficient catalyst as well as solvent by conventional method in short reaction times with excellent yield. The newly synthesized benzimidazole and benzoxazole derivatives were evaluated for antimicrobial and antioxidant activity and exhibited excellent to good activities compared to the standard drugs. Furthermore, the theoretical predictions based on molecular docking against microbial DNA *gyrase* could provide an insight into the plausible mechanism of action and establish a link between the observed antimicrobial activity and the binding affinity shedding light on specific thermodynamic (bonded and nonbonded) interactions governing the activity. Furthermore, the synthesized compounds were analyzed for absorption, distribution, metabolism, and excretion properties and exhibited potential properties to build up as good oral drug candidates.

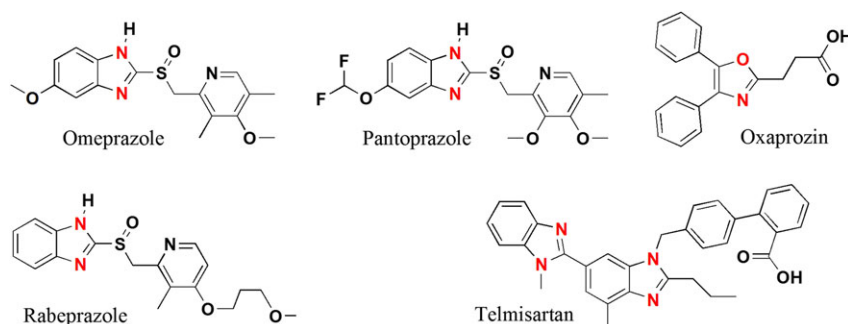
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## INTRODUCTION

The benzimidazole and benzoxazole moieties are an important heterocyclic nucleus which has been used extensively in medicinal chemistry. Benzimidazole is a component of vitamin B12 and is related to the DNA base purine and the stimulant caffeine. Substituted benzimidazole displays a broad spectrum of potential pharmacological activities and is present in a number of pharmacologically active molecules such as bendamustine, omeprazole, pantoprazole, lansoprazole, oxaprozin, rabeprazole, telmisartan, alendazole, mebendazole, and thiabendazole (Fig. 1). The discovery of these classes of drugs having the benzimidazole and benzoxazole structure provides an outstanding case history of modern drug development and also points out the unpredictability of biological activities

from structural modification of a prototype drug molecules having a variety of medicinal applications. The substituted benzimidazole and benzoxazole structures were associated with a wide range of biological activities including anticancer, antiviral, antibacterial, antifungal, antihelminthic, anti-inflammatory, antihistaminic, proton pump inhibitor, antioxidant, antihypertensive, and anticoagulant activities. Their derivatives were also found to exhibit cytotoxic activity. Furthermore, some of them were found to be fluorescent whitening agents.

Considering the importance of benzimidazole and benzoxazole in biological systems, it was thought that it would be worthwhile to design the syntheses of compounds containing both a series of new benzimidazole and benzoxazole derivatives and screen them for antimicrobial and antioxidant activities.



**Figure 1.** Structures of biologically active Active Pharmaceutical Ingredient (API) molecules. [Color figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

Various methods have been reported for the synthesis of these heterocycles including the condensation of carboxylic acids, orthoesters, acid chlorides, nitriles, amides, aldehydes, and esters with *o*-substituted amino aromatics derivatives from orthoesters. Generally, in the pharmaceutical field, new drugs are continuously discovered by molecular modification of lead compound of established activity. Molecular modification can possibly result in augmenting the activity. Combination of separate groups having similar activity in one compound by eliminating, substituting, or adding new moiety to parent lead compound gives molecular modification. In the survey of literature, it is seen that drug design by molecular modification is a productive source of new drugs; therefore, there is a need to synthesize new molecules as potential medicinal agents. Among medicinal agents, there is growing interest in the development of newer effective antimicrobial and antioxidant agents.

Among the variety of compounds studied, benzimidazole and benzoxazole derivatives form an important class. Mainly there are two general methods for synthesizing 2-substituted benzoxazoles, benzimidazoles, and benzothiazoles. One is the coupling of *o*-substituted amino aromatics with carboxylic acid derivatives and acyl chlorides, which is either catalyzed by strong acids or microwave conditions. The other is the oxidative cyclization of phenolic Schiff bases derived from the condensation of *o*-substituted amino aromatics and aldehydes. In latter reactions, various oxidants have been used. Different catalysts and methods were also reported for the preparations of these heterocycles. Synthesis of substituted benzoxazoles, benzimidazoles, and benzothiazoles by using palladium and 2,3-dichloro-5,6-dicyano-1,4-benzoquinone-mediated oxidative cyclization [1], 1,4-diazabicyclo[2.2.2]octane [2],  $\text{ZrOCl}_2 \cdot 8\text{H}_2\text{O}$  [3], silica sulfuric acid [4], Indion 190 resin [5], ruthenium-catalyzed hydrogen-transfer reactions for the conversion of alcohols into benzimidazoles and aldehydes into benzoxazoles and benzothiazoles [6], ionic liquids [7], resin bound, phenyliodonium bis(trifluoroacetate) and

microwave irradiation [8], methanesulphonic acid [9], copper (II) catalyzed [10], pyridinium chlorochromate [11], activated carbon (Darco KB) [12],  $\text{In}(\text{OTf})_3$  [13], tin (II) chloride [14],  $\text{BF}_3 \cdot \text{OEt}_2$  [15], polyphosphoric acid (PPA) [16], ceric ammonium nitrate [17], hypervalent iodine catalytic amount of iodine [18],  $\text{SiO}_2/\text{ZnCl}_2$  [19], PEG-mediated catalyst free [20], thionyl chloride and  $\text{HCl}$  [21], catalytic amount of  $\text{K}_2\text{CO}_3$  in NMP [22], baker's yeast [23],  $\text{H}_2\text{O}_2/\text{Fe}(\text{NO}_3)_3$  [24], PTSA,  $\text{NaHSO}_4\text{-SiO}_2$ , and cetyltrimethyl ammonium bromide [25]. However, many of these methodologies suffer from one or more disadvantages, such as low yields, lack of easy availability of the starting materials, prolonged reaction time, high temperature, expensive catalysts, special apparatus, and harsh reaction conditions. Thus, there is a need for simple and efficient processes for the synthesis of benzoxazoles, benzimidazoles, and benzothiazoles derivatives. Selected catalyst as well as solvent for reaction, that is, PPA, is commercially available, cheap, and more active catalyst than many reported expensive catalysts.

## RESULT AND DISCUSSION

In recent years, the use of microwave irradiation to simplify and improve classical organic reaction has become a very popular method because it often leads to high yield, clean reaction, and shorter reaction time. Synthetic methods are disclosed for synthesis of 2-substituted benzimidazole and benzoxazole derivatives like microwave irradiation conditions and classical conditions [16c]. To study the actual role of PPA as catalyst and solvent, we performed the combinational study with different solvents like DCM,  $\text{CH}_3\text{CN}$ , THF,  $\text{CHCl}_3$ , DMSO, DMF, MeOH, EtOH, *n*-butanol, PEG 400, and dioxane by keeping PPA (1%) as catalyst constant obtained a very low yield (Table 1). These results suggest that plane PPA is the best catalyst as well as solvent for synthesis of 2-substituted benzimidazole and benzoxazole derivatives, because of its acidic nature

Table 1

Screening of different solvents and their effect on reaction time and yield for the synthesis of 2-substituted benzimidazole and benzoxazole derivatives.

| Sr. no. | Solvent           | Temperature (°C) | Time (h) | Yield (%) |
|---------|-------------------|------------------|----------|-----------|
| 1       | DCM               | 40               | 24       | -         |
| 2       | ACN               | 85               | 18       | 35        |
| 3       | THF               | 68               | 19       | 20        |
| 4       | CHCl <sub>3</sub> | 82               | 24       | -         |
| 5       | DMSO              | 180              | 13       | 65        |
| 6       | DMF               | 155              | 15       | 60        |
| 7       | MeOH              | 65               | 18.5     | 56        |
| 8       | EtOH              | 80               | 16.5     | 63        |
| 9       | n-Butanol         | 115              | 19       | 58        |
| 10      | PEG-400           | 180              | 15       | 72        |
| 11      | Dioxane           | 180              | 12.5     | 70        |
| 12      | PPA               | 145              | 4.5      | 95        |

<sup>a</sup>Reaction conditions: *N*-methyl-*o*-phenylenediamine hydrochloride/2-amino phenol (1.05 mmol), aromatic aldehyde/acid (1 mmol), and catalytic PPA (1%) were stirred in solvent (3 vol) or neat.

<sup>b</sup>Isolated yields.

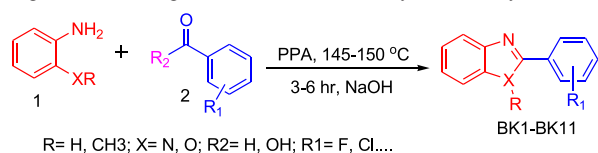
and greater solubility and dehydration affinity to cyclize the compound than in organic solvent.

Herein, we have carried out the reaction of various substituted aromatic aldehyde/acid with *N*-methyl-*o*-phenylenediamine hydrochloride/2-amino-phenol in the presence of PPA at 145–150°C for 3–6 h by conventional method obtaining products (**BK-1–11**) in very good to excellent yields in shorter reaction time (Scheme 1) and for plausible reaction mechanism (Schemes 2 and 3).

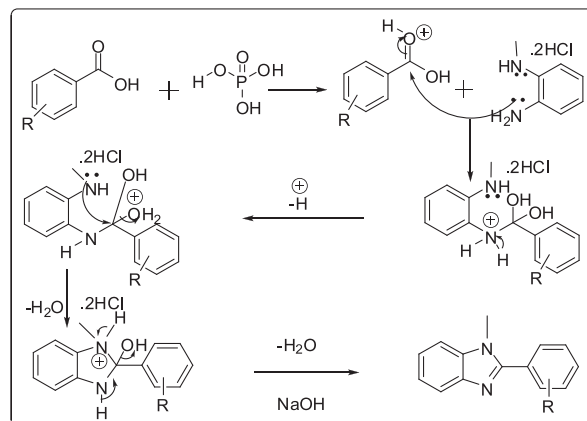
All the reactions were completed within 3–6 h with 80–95% yield. All the products were confirmed by <sup>1</sup>H-NMR, <sup>13</sup>C-NMR, IR, and mass spectroscopy (Supporting Information). In the FTIR, the diagnostic bands of benzimidazole and benzoxazole ring at 1650–1580, 1580–1400, and 1200–1000 cm<sup>-1</sup> were assigned to C=N, C=C, C-N, and C-O stretch, respectively. In the <sup>13</sup>C-NMR spectra, the peaks at 152–148, 144–138, and 33–30 ppm were assigned to C=N, C-N, and C-N-CH<sub>3</sub> of benzimidazole ring, respectively. In the <sup>1</sup>H-NMR spectra, the peaks at 3.85–3.70 ppm appeared as a singlet assigned to the 3H of the -N-CH<sub>3</sub> of benzimidazole ring.

We have tried alternative process [21b] for the synthesis of 2-substituted benzimidazole and benzoxazole

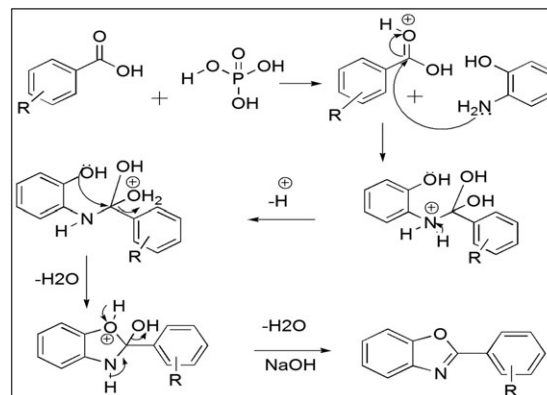
**Scheme 1.** Synthesis of 2-substituted benzimidazole and benzoxazole using PPA. [Color figure can be viewed at wileyonlinelibrary.com]



**Scheme 2.** Plausible reaction mechanism of the desired benzimidazole.



**Scheme 3.** Plausible reaction mechanism of the desired benzoxazoles.



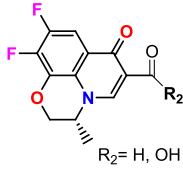
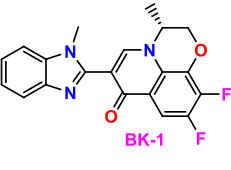
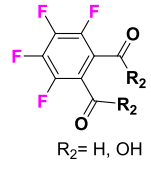
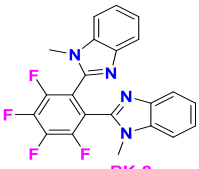
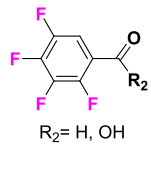
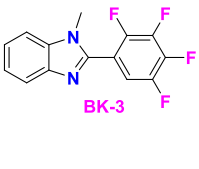
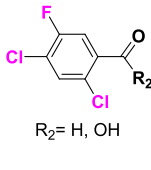
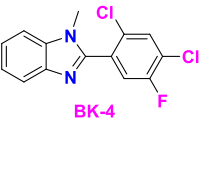
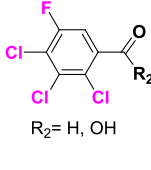
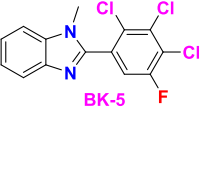
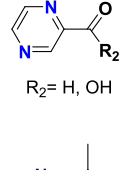
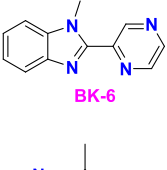
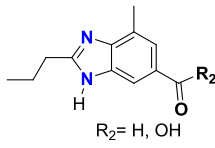
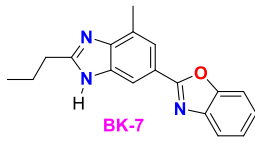
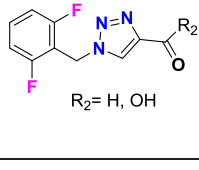
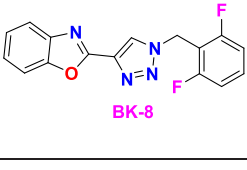
derivatives by using Eaton's reagent mix of P<sub>2</sub>O<sub>5</sub>:MSA (1:10) 7 vol and *N*-methyl-*o*-phenylenediamine hydrochloride/2-amino phenol (1.05 mmol); aromatic aldehyde or acid (1 mmol) was stirred at 100°C for 12 h, but the reaction does not comply over thin-layer chromatography (TLC), and isolated yield was very poor (45–50%). Even though in 4N to 10N HCl (7 vol) and *o*-phenylenediamine/2-amino phenol (1.05 mmol), aromatic aldehyde or acid (1 mmol) were refluxed up to 20 h reaction that not complies over TLC and isolated yield (35–40%), and results are very disappointing as compared to reaction in PPA yield (80–95%, Table 2).

**Antibacterial activity.** The results of the antibacterial activity screening of the tested compound on gram-positive and gram-negative bacteria are summarized in Table 3 and Figure 2.

From the minimum inhibitory concentration (MIC) data, it is clear that the presence of 9,10-difluoro, imidazole, [1,4]oxazino, quinolin-7-one moieties in **BK-1**, 2-substituted benzoxazole having substituted benzimidazole in **BK-7** and 2-substituted benzoxazole having 2,3,4-trichloro-5-fluorophenyl moiety in **BK-10** may enhanced

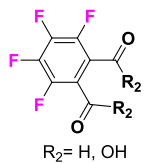
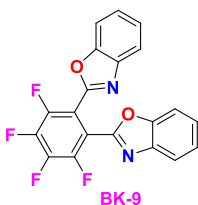
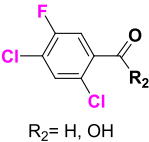
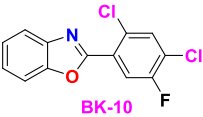
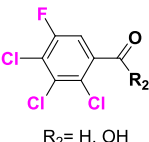
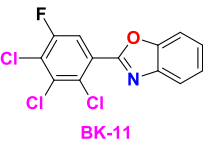
**Table 2**

2-Substituted benzimidazoles and benzoxazoles with yield and physical data.

| Comp | Aromatic aldehyde/acid                                                                                        | Product                                                                                     | Yield (%) | Melting point (°C) |
|------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------|--------------------|
| BK-1 | <br>R <sub>2</sub> = H, OH   | <br>BK-1   | 83        | 240–244            |
| BK-2 | <br>R <sub>2</sub> = H, OH   | <br>BK-2   | 84        | >260               |
| BK-3 | <br>R <sub>2</sub> = H, OH   | <br>BK-3   | 92        | >260               |
| BK-4 | <br>R <sub>2</sub> = H, OH  | <br>BK-4  | 88        | 230–233            |
| BK-5 | <br>R <sub>2</sub> = H, OH | <br>BK-5 | 89        | 235–238            |
| BK-6 | <br>R <sub>2</sub> = H, OH | <br>BK-6 | 95        | >260               |
| BK-7 | <br>R <sub>2</sub> = H, OH | <br>BK-7 | 85        | 225–227            |
| BK-8 | <br>R <sub>2</sub> = H, OH | <br>BK-8 | 88        | 254–256            |

(Continues)

**Table 2**  
(Continued)

| Comp         | Aromatic aldehyde/acid                                                                                      | Product                                                                                    | Yield (%) | Melting point (°C) |
|--------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------|--------------------|
| <b>BK-9</b>  | <br>R <sub>2</sub> = H, OH | <br>BK-9  | 95        | 242–246            |
| <b>BK-10</b> | <br>R <sub>2</sub> = H, OH | <br>BK-10 | 89        | 220–222            |
| <b>BK-11</b> | <br>R <sub>2</sub> = H, OH | <br>BK-11 | 94        | >260               |

<sup>a</sup>Reaction conditions: *N*-methyl-*o*-phenylenediamine hydrochloride/2-amino phenol (1.05 mmol) and aromatic aldehyde/acid (1 mmol), namely, R<sub>2</sub> = H, OH, and catalytic PPA (1%), were stirred in solvent (3 vol) or neat.

<sup>b</sup>Isolated yields.

**Table 3**

*In vitro* antimicrobial and antioxidant evaluation MIC values (μg/mL) of compounds (**BK-1–11**).

| Compounds     | Antibacterial activity |                  |                |                      | Antifungal activity |                  |                     | Antioxidant activity |       |
|---------------|------------------------|------------------|----------------|----------------------|---------------------|------------------|---------------------|----------------------|-------|
|               | <i>B. subtilis</i>     | <i>S. aureus</i> | <i>E. coli</i> | <i>P. aeruginosa</i> | <i>A. niger</i>     | <i>A. flavus</i> | <i>M. purpureus</i> | DPPH                 | NO    |
| <b>BK-1</b>   | 1.25                   | 1.25             | 0.625          | 1.25                 | 12.5                | 25.0             | 6.25                | 32.07                | 42.10 |
| <b>BK-2</b>   | 2.5                    | 1.25             | 2.5            | 5.0                  | 25.0                | 75.0             | 50.0                | 44.34                | 58.36 |
| <b>BK-3</b>   | 5.0                    | 5.0              | 5.0            | 5.0                  | 50.0                | 50.0             | 75.0                | 38.25                | 61.45 |
| <b>BK-4</b>   | 2.5                    | 5.0              | 5.0            | 5.0                  | 100                 | 125              | NA                  | 35.17                | 76.54 |
| <b>BK-5</b>   | 2.5                    | 1.25             | 2.5            | 2.5                  | NA                  | 150              | 125                 | 44.54                | 81.45 |
| <b>BK-6</b>   | 5.0                    | 5.0              | 5.0            | 5.0                  | 25.0                | NA               | 12.5                | 22.45                | 46.10 |
| <b>BK-7</b>   | 0.156                  | 0.625            | 1.25           | 1.25                 | 12.5                | 12.5             | 6.25                | 27.52                | 40.36 |
| <b>BK-8</b>   | 5.0                    | 5.0              | 2.5            | 5.0                  | 25.0                | 50.0             | 25.0                | 20.56                | 48.56 |
| <b>BK-9</b>   | 1.25                   | 2.5              | 2.5            | 5.0                  | 50.0                | 75.0             | 50.0                | 50.62                | 52.10 |
| <b>BK-10</b>  | 2.5                    | 2.5              | 0.156          | 0.312                | 75.0                | 100              | 125                 | 63.85                | 88.10 |
| <b>BK-11</b>  | 2.5                    | 2.5              | 1.25           | 1.25                 | 50                  | 75               | 100                 | 58.35                | 85.64 |
| Ciprofloxacin | 3.23                   | 3.23             | 1.57           | 1.67                 | -                   | -                | -                   | -                    | -     |
| Miconazole    | -                      | -                | -              | -                    | 25                  | 25               | 12.5                | -                    | -     |
| Fluconazole   | -                      | -                | -              | -                    | 12.5                | 12.5             | 6.25                | -                    | -     |
| Ascorbic acid | -                      | -                | -              | -                    | -                   | -                | -                   | 42.76                | 44.93 |

MIC, minimum inhibitory concentration; NA, no activity was observed up to 100 μg/mL of bacterial strains (*Bacillus subtilis*, *Staphylococcus aureus*, *Escherichia coli*, and *Pseudomonas aeruginosa*) and fungal strains (*Aspergillusniger*, *Aspergillus flavus*, and *Monascus purpureus*) for antioxidant 2,2-diphenyl-1-picrylhydrazyl (DPPH) and nitric oxide radical scavenging.

the more potency broad spectrum antibacterial substances, whereas **BK-2**, **BK-5**, **BK-9**, and **BK-11** are less potent broad spectrum antibacterial substances. However, some compounds **BK-3**, **BK-4**, **BK-6**, and **BK-8** were found to have very poor activity against screening on gram-positive and gram-negative bacteria.

**Antifungal activity.** The minimum inhibitory potential of newly synthesized 2-substituted benzimidazole and benzoxazole derivatives **BK-1–11** was evaluated *in vitro* against three different fungal strains: *Aspergillusniger*, *Aspergillusflavus*, and *Monascuspurpureus*; and results were compared with standard drug miconazole and

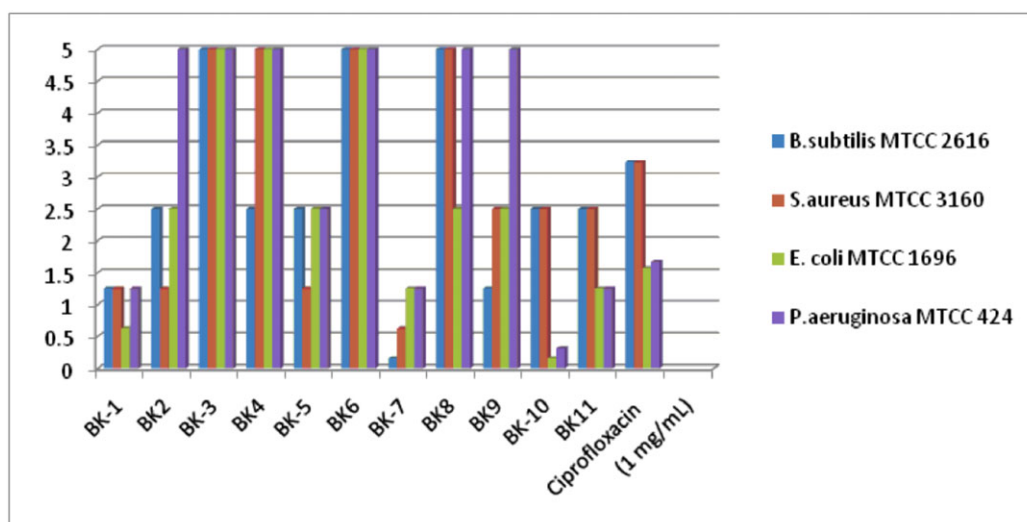


Figure 2. *In vitro* antibacterial activity of series of compound BK-1–11. [Color figure can be viewed at wileyonlinelibrary.com]

fluconazole. The MIC values in  $\mu\text{g/mL}$  were estimated, and the results are summarized in Table 3. Among all the synthesized compounds, **BK-7** having benzimidazole and benzoxazole moieties and **BK-1** having benzimidazole with ofloxacin moieties in a single molecular framework showing good activity as compared to standard drugs miconazole and fluconazole. Remaining compounds other than **BK-1** and **BK-7** show moderate antifungal activity compared to the fungal strains *Aspergillus niger*, *Aspergillus flavus*, and *Monascus purpureus* compared to the standard drugs.

**Antioxidant activity.** The results of the antioxidant activity screening of the tested compound for DPPH and nitric oxide (NO) radical scavenging activities were also summarized in Table 3 and Figure 3. 2-Substituted

benzimidazole and benzoxazole derivatives **BK-1** to **BK-11** shows good to moderate antioxidant activity as compared to the standard drug ascorbic acid (Table 3). According to the DPPH assay, compounds **BK-8**, **BK-6**, **BK-7**, **BK-1**, **BK-4**, and **BK-3** exhibited promising radical scavenging activities when compared with synthetic antioxidant ascorbic acid ( $\text{IC}_{50} = 42.76 \mu\text{g/mL}$ ), with  $\text{IC}_{50}$  values of 20.56, 22.45, 27.52, 32.07, 35.17, and  $38.25 \mu\text{g/mL}$ , respectively. From the synthesized compounds, compounds **BK-8**, **BK-6**, and **BK-7** showed the best DPPH radical scavenging activity, while compound **BK-10** showed the lowest activity when compared with ascorbic acid. From the MIC data, it is clear that the presence of tetra and tri-halogenated aromatic substitution at 2 position on benzoxazole is an

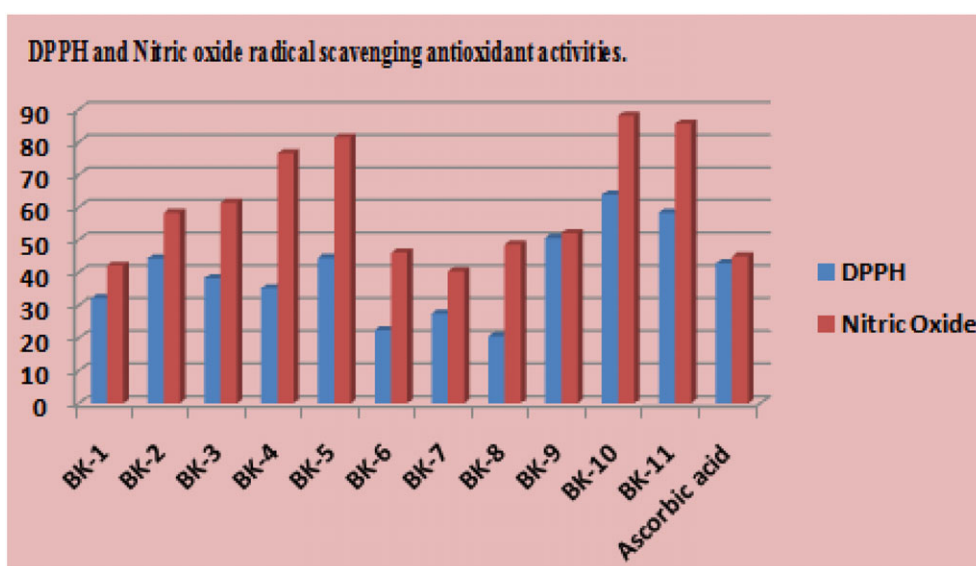


Figure 3. *In vitro* antioxidant activity of series of compound BK-1 to BK-11. [Color figure can be viewed at wileyonlinelibrary.com]

excellent potent as compared to others for DPPH radical scavenging activity, benzoxazole having 2,3,4-trichloro-5-fluorophenyl moiety in **BK-11**, **BK-10**, and **BK-9** may enhanced the more potency antioxidant, whereas all other are good to poor antioxidant.

Except compounds **BK-1** and **BK-7**, all other synthesized compounds show very poor antioxidant activity for NO radical scavenging activity compared to reference standard drug ascorbic acid.

**Molecular docking.** The level of antibacterial activities demonstrated by the title compounds prompted us to perform molecular docking studies to understand the ligand-protein interactions. *In silico* approach, molecular docking has become an integral part of the drug discovery process to predict, with a substantial level of accuracy, the conformation of ligands within the target active site and also imparts knowledge on binding affinities and the nature of the thermodynamic interaction (van der Waals, hydrogen bonding, electrostatic) between the drug and its target that govern the observed biological activity. Bacterial DNA gyrase is required for DNA synthesis. Inhibition of DNA gyrase blocks relaxation of supercoiled DNA, relaxation being a requirement for transcription and replication. Owing to a critical role in bacterial cell survival, DNA gyrase has been exploited as a validated drug target for antibacterial chemotherapy. Furthermore, it is essential enzyme in all bacteria (prokaryotes), but its absence in higher eukaryotes makes it an attractive target for antibacterial chemotherapy [26]. Therefore, with the aim of gaining an insight into plausible mechanism for antimicrobial activities demonstrated by novel benzimidazole and benzoxazole derivatives discussed herein, a molecular docking study was performed against *DNA gyrase subunit b* (pdb id:1KZN). All the synthesized molecules (**BK-1** to **BK-11**) showed good binding affinity with docking scores ranging from  $-8.465$  to  $-7.00$  and binding energies ranging from  $-49.642$  to  $-39.176$  kcal/mol with each of these molecules engaging in a series of steric and electrostatic interactions with the amino acid residues lining the active site of the target enzyme. A detailed investigation of the per-residue bonded and nonbonded interactions between these molecules and the active site residues of the enzyme was carried out to identify the most significantly interacting residues and also their nature of interactions which assist in the lead optimization (Table 4). This analysis is being elaborated for one of the most active compound (**BK-7**) in the next section.

The most stable docked conformation of **BK-7** is oriented into the active site of bacterial DNA gyrase site engaging in a network of steric and electrostatic interactions, and binding is stabilized furthermore by a hydrogen bond and pi-pi stacking interaction (Fig. 4). The most prominent steric interactions were observed

through the benzimidazole nucleus with Val167 ( $-1.983$  kcal/mol), Thr165 ( $-2.684$  kcal/mol), Val120 ( $-1.917$  kcal/mol), Ala96 ( $-1.999$  kcal/mol), His95 ( $-1.887$  kcal/mol), Ile90 ( $-1.991$  kcal/mol), Asp73 ( $-2.593$  kcal/mol), Val71 ( $-1.969$  kcal/mol), Glu50 ( $-3.793$  kcal/mol), Ala47 ( $-2.588$  kcal/mol), Asn46 ( $-4.229$  kcal/mol), and Val43 ( $-1.811$  kcal/mol) while the benzoxazole nucleus was observed to engaged in similar interactions with Gly119 ( $-1.989$  kcal/mol), Pro79 ( $-2.36$  kcal/mol), Ile78 ( $-4.506$  kcal/mol), Gly77 ( $-1.981$  kcal/mol), Arg76 ( $-2.929$  kcal/mol), and Asp49 ( $-1.887$  kcal/mol) residues of the active site. The high binding affinity of **BK7** is also attributed to a relatively few but significant electrostatic interactions observed with Thr165 ( $-1.886$  kcal/mol), Gly77 ( $-1.455$  kcal/mol), Asp73 ( $-1.493$  kcal/mol), Glu50 ( $-1.924$  kcal/mol), and Asn46 ( $-1.165$  kcal/mol) residues in the active site of DNA gyrase. The hydrogen atom attached to nitrogen (-NH-) of the benzimidazole scaffold formed one hydrogen bond with the oxygen atom of the Asp73 having a bond length of  $1.85$  Å.

Furthermore, a very prominent pi-pi stacking interaction was also observed between the Arg76 residue of the active site and the benzene ring of the benzoxazole nucleus ( $2.046$  Å). These hydrogen bonding and pi-pi stacking interactions anchor the ligand into the active site and facilitate the nonbonded interactions (steric and electrostatic) between the molecule and the enzyme. The 3D model of the interaction between **BK-7** and the DNA gyrase binding site is depicted in Figure 4, showing well binding affinity to the target. The results of the molecular docking study show that the benzimidazole and benzoxazole nucleus act synergistically to interact with the DNA gyrase binding site, suggesting that compound **BK-7** is a potential inhibitor of this enzyme. A similar network of well-balanced steric and electrostatic nonbonded interactions and bonded interactions was also observed for other molecules in the series (Figs S1-S10). This binding affinity data and more specifically the per-residue ligand interaction analysis will be fruitfully utilized for the structure-based lead optimization of these benzimidazole and benzoxazole derivatives to arrive at potent antibacterial agents.

**In silico absorption, distribution, metabolism, and excretion properties.** The success of a drug is determined by good efficacy and acceptable absorption, distribution, metabolism, and excretion (ADME) profile. In the present study, drug-likeness model score (a collective property of physic-chemical properties, pharmacokinetics, and pharmacodynamics of a compound is represented by a numerical value) was computed by MolSoft software [27]. A computational study of all the synthesized compounds was performed for prediction of ADME properties, and the value obtained is presented in

**Table 4**

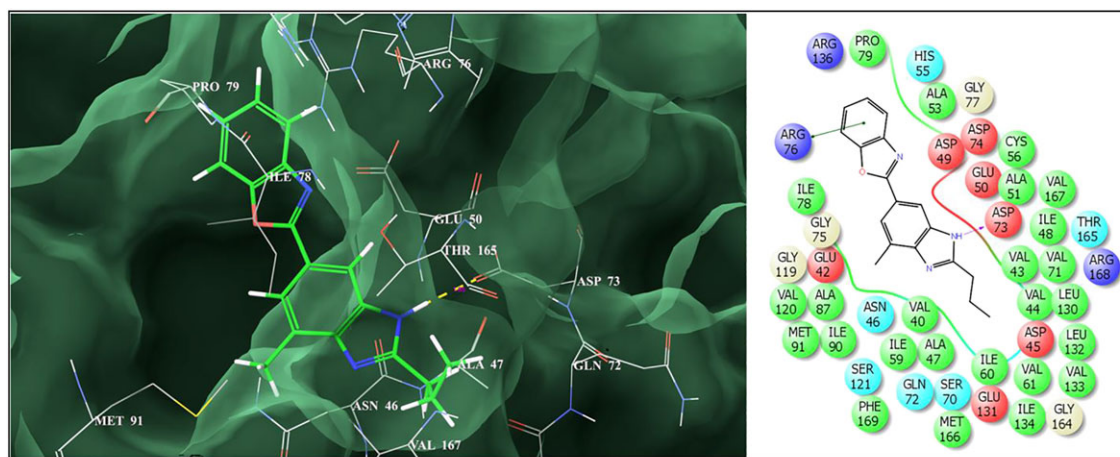
The per-residue interaction analysis of the molecular docking study of 2-substituted benzimidazole and benzoxazole derivatives on active site of *DNA gyrase subunit b*.

| Comp        | Docking score | Glide interaction energy (kcal/mol) | Per-residues interactions                                                                                                                                                                                                                                                                         |                                                                                 |              |                    |
|-------------|---------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------|--------------------|
|             |               |                                     | Van der Waals (kcal/mol)                                                                                                                                                                                                                                                                          | Coulombic (kcal/mol)                                                            | H-bonds (Å)  | Pi-pi stacking (Å) |
| <b>BK-1</b> | −8.333        | −48.364                             | Val167 (−1.805), Thr165 (−2.671), Val120 (−1.842), Gly119 (−1.778), Ala96 (−1.899), His95 (−1.878), Ile90 (−1.262), Pro79 (−2.958), Ile78 (−4.278), Gly77 (−1.978), Arg76 (−2.559), Asp73 (−2.133), Val71 (−1.556), Glu50 (−3.786), Asp49 (−1.366), Ala47 (−2.264), Asn46 (−5.103), Val43 (−1.42) | Glu50 (−1.853), Asn46 (−2.015)                                                  | Asn46 (1.85) |                    |
| <b>BK-2</b> | −7.885        | −44.339                             | Val167 (−1.667), Thr165 (−1.604), Val120 (−1.334), Gly119 (−1.556), Ala96 (−1.815), His95 (−1.067), Ile90 (−2.579), Pro79 (−2.227), Ile78 (−3.671), Gly77 (−1.579), Arg76 (−2.292), Asp73 (−2.126), Glu50 (−2.547), Asp49 (−1.747), Asn46 (−3.155)                                                | Thr165 (−1.082), Arg136 (−1.622), Arg76 (−1.20), Asn46 (−4.355)                 | Asn46 (2.03) |                    |
| <b>BK-3</b> | −7.000        | −39.176                             | Val167 (−1.434), Thr165 (−1.195), Val120 (−1.134), Gly119 (−1.136), Ala96 (−1.788), His95 (−1.156), Ile90 (−2.414), Ile78 (−4.211), Gly77 (−1.405), Arg76 (−2.882), Asp73 (−1.517), Val71 (−1.1), Glu50 (−3.643), Ala47 (−2.288), Asn46 (−4.801), Val43 (−1.876)                                  | Asp73 (−6.663), Glu50 (−2.229), Asn46 (−1.06)                                   | -            | -                  |
| <b>BK-4</b> | −7.115        | −41.809                             | Val167 (−1.555), Thr165 (−1.703), Val120 (−1.995), Gly119 (−1.539), Ala96 (−1.791), His95 (−1.178), Ile90 (−2.443), Pro79 (−2.290), Ile78 (−3.359), Gly77 (−1.702), Arg76 (−1.925), Val71 (−1.447), Glu50 (−3.501), Ala47 (−2.029), Asn46 (−5.275), Val43 (−1.631)                                | Thr165 (−1.779), Asp73 (−2.611), Glu50 (−1.943), Asn46 (−1.681)                 | Asn46 (2.76) |                    |
| <b>BK-5</b> | −7.979        | −45.804                             | Val167 (−1.736), Thr165 (−2.218), Val120 (−1.337), Gly119 (−1.626), Ala96 (−1.847), His95 (−1.074), Ile90 (−2.638), Pro79 (−2.204), Ile78 (−3.491), Gly77 (−1.679), Arg76 (−2.243), Asp73 (−2.165), Val71 (−1.99), Glu50 (−2.895), Asn46 (−3.592)                                                 | Thr165 (−1.877), Gly77 (−1.461), Asp73 (−1.493), Glu50 (−1.911), Asn46 (−1.561) | Asn46 (2.12) | -                  |
| <b>BK-6</b> | −7.020        | −40.661                             | Val167 (−1.504), Thr165 (−1.644), Val120 (−1.479), Gly119 (−1.465), Ala96 (−1.756), His95 (−1.148), Ile90 (−2.307), Ile78 (−3.68), Asp73 (−2.416), Glu50 (−1.315), Ala47 (−1.69), Asn46 (−4.068), Val43 (−1.856)                                                                                  | Asp73 (−3.831), Glu50 (−2.724), Asn46 (−2.171)                                  | Asn46 (2.44) |                    |
| <b>BK-7</b> | −8.465        | −49.642                             | Val167 (−1.983), Thr165 (−2.684), Val120 (−1.917), Gly119 (−1.989), Ala96 (−1.999), His95 (−1.887), Ile90 (−1.991), Pro79 (−2.36), Ile78 (−4.506), Gly77 (−1.981), Arg76 (−2.929), Asp73 (−2.593), Val71 (−1.969), Glu50 (−3.793), Asp49 (−1.887), Ala47 (−2.588), Asn46 (−4.229), Val43 (−1.811) | Thr165 (−1.886), Gly77 (−1.455), Asp73 (−1.493), Glu50 (−1.924), Asn46 (−1.165) | Asp73 (1.85) | Arg76              |

(Continues)

**Table 4**  
(Continued)

| Comp         | Docking score | Glide interaction energy (kcal/mol) | Per-residues interactions                                                                                                                                                                                                                          |                                                                                 |              |                    |
|--------------|---------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------|--------------------|
|              |               |                                     | Van der Waals (kcal/mol)                                                                                                                                                                                                                           | Coulombic (kcal/mol)                                                            | H-bonds (Å)  | Pi-pi stacking (Å) |
| <b>BK-8</b>  | -7.219        | -42.080                             | Val167 (-1.629), Thr165 (-1.457), Val120 (-2.087), Gly119 (-1.541), Ala96 (-1.799), His95 (-1.055), Ile90 (-2.531), Pro79 (-2.303), Ile78 (-3.635), Gly77 (-1.798), Val71 (-1.857), Glu50 (-1.855), Ala47 (-2.059), Asn46 (-5.034), Val43 (-1.864) | Thr165 (-1.798), Gly77 (-1.438), Asp73 (-2.776), Glu50 (-3.601)                 | --           | --                 |
| <b>BK-9</b>  | -7.389        | -43.326                             | Val167 (-1.646), Thr165 (-1.557), Val120 (-1.330), Gly119 (-1.543), Ala96 (-1.809), His95 (-1.065), Ile90 (-2.586), Pro79 (-2.315), Ile78 (-4.113), Gly77 (-1.801), Arg76 (-2.443), Asp73 (-1.903), Glu50 (-3.126), Asn46 (-3.39)                  | Thr165 (-1.757), Arg136 (-1.027), Arg76 (-1.075), Asn46 (-1.178)                | Asn46 (2.40) | Arg76              |
| <b>BK-10</b> | -8.116        | -46.294                             | Val167 (-1.764), Thr165 (-2.545), Val120 (-1.345), Ile90 (-1.228), Pro79 (-2.819), Ile78 (-4.197), Gly77 (-1.889), Arg76 (-2.423), Asp73 (-2.158), Glu50 (-3.295), Asp49 (-1.333), Ala47 (-2.146), Asn46 (-4.791), Val43 (-1.766)                  | Thr165 (-1.768), Gly77 (-1.245), Asp73 (-3.93), Glu50 (-2.313), Asn46 (-1.155)  | --           | --                 |
| <b>BK-11</b> | -8.110        | -46.544                             | Val167 (-1.738), Thr165 (-2.226), Val120 (-1.333), Ile90 (-1.226), Pro79 (-2.743), Ile78 (-3.533), Gly77 (-1.777), Arg76 (-2.435), Asp73 (-2.176), Val71 (-1.99), Glu50 (-3.245), Ala47 (-2.121), Asn46 (-4.267), Val43 (-1.723)                   | Arg136 (-1.497), Arg76 (-1.196), Asp73 (-3.456), Glu50 (-2.221), Asn46 (-1.151) | -            | -                  |



**Figure 4.** Binding mode of **BK-7** into the active site of *DNA gyrase subunit b* (on right side: green lines signify pi-pi stacking interactions while the pink lines represent the hydrogen bonding interactions). [Color figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

Table 4. Absorption (% ABS) was calculated by: % ABS =  $109 - (0.345 \times \text{TPSA})$  [28]. It is observed that compounds exhibited a good absorption (% ABS) ranging from 89.42% to 102.84%. Furthermore, none of

the synthesized compounds **BK-1** to **BK-11** violated Lipinski's rule of five ( $\text{miLog } P \leq 5$ ). A molecule likely to be developed as an orally active drug candidate should not show more than one violation of the following four

Table 5

Pharmacokinetic parameters important for good oral bioavailability and drug likeness model score of the synthesized compounds **BK-1–11**.

| Entry        | % ABS  | TPSA (Å <sup>2</sup> ) | n-ROTB | MV     | MW     | miLog <i>P</i> | n-ON | n-OHNNH | Lipinski violation | Drug likeness model score |
|--------------|--------|------------------------|--------|--------|--------|----------------|------|---------|--------------------|---------------------------|
| Rule         | -      | -                      | -      | -      | <500   | ≤5             | <10  | <5      | ≤1                 |                           |
| <b>BK-1</b>  | 92.07  | 49.06                  | 1      | 303.75 | 367.36 | 2.01           | 5    | 0       | 0                  | 0.16                      |
| <b>BK-2</b>  | 96.70  | 35.65                  | 2      | 330.11 | 410.37 | 5.66           | 4    | 0       | 1                  | -0.99                     |
| <b>BK-3</b>  | 102.84 | 17.83                  | 1      | 216.94 | 280.22 | 4.05           | 2    | 0       | 0                  | -0.74                     |
| <b>BK-4</b>  | 102.84 | 17.83                  | 1      | 229.22 | 295.14 | 4.99           | 2    | 0       | 0                  | -0.95                     |
| <b>BK-5</b>  | 102.84 | 17.83                  | 1      | 242.75 | 329.59 | 5.59           | 2    | 0       | 1                  | -0.75                     |
| <b>BK-6</b>  | 93.95  | 43.61                  | 1      | 188.90 | 210.24 | 1.78           | 4    | 0       | 0                  | -0.64                     |
| <b>BK-7</b>  | 90.12  | 54.72                  | 3      | 268.40 | 291.35 | 4.66           | 4    | 1       | 0                  | -0.24                     |
| <b>BK-8</b>  | 89.42  | 56.75                  | 3      | 251.96 | 312.28 | 3.27           | 5    | 0       | 0                  | -0.21                     |
| <b>BK-9</b>  | 91.03  | 52.06                  | 2      | 289.39 | 384.29 | 5.72           | 4    | 0       | 1                  | -1.24                     |
| <b>BK-10</b> | 100.01 | 26.03                  | 1      | 208.86 | 282.10 | 5.02           | 2    | 0       | 1                  | -1.12                     |
| <b>BK-11</b> | 100.01 | 26.03                  | 1      | 222.39 | 316.55 | 5.63           | 2    | 0       | 1                  | -0.91                     |

% ABS, percentage absorption; miLog *P*, logarithm of partition coefficient of compound between n-octanol and water; MV, molecular volume; MW, molecular weight; n-OHNNH donors, number of hydrogen bonds donors; n-ON acceptors, number of hydrogen bond acceptors; n-ROTB, number of rotatable bonds; TPSA, topological polar surface area.

criteria: miLog *P* (octanol–water partition coefficient) ≤5, molecular weight ≤500, number of hydrogen bond acceptors ≤10, and number of hydrogen bond donors ≤5 [29].

The larger the value of the drug likeness model score is, the higher the probability that the particular molecule will be active. All the tested compounds **BK-1–11** followed the criteria for orally active drug (Table 5), and therefore, these compounds may have a good potential for eventual development as oral agents.

## CONCLUSIONS

The preliminary *in vitro* antibacterial, antifungal, and antioxidant screening results of novel 2-substituted benzimidazole and benzoxazole derivatives reported here have emerged as highly potent antibacterial and antioxidant agents. The possible improvements in the activity can be further slight modifications in the substituents on the basic benzimidazole and benzoxazole nucleus. The tetra and tri-substituted fluorine, chlorine on aromatic ring and hybrid molecule like (**BK-7**) having both moieties benzimidazole and benzoxazole has highly increased the activity of the compounds compared to other substituent on aromatic ring which are present on novel 2-substituted benzimidazole and benzoxazole derivatives. But all the other compound showing very poor activity against screening of fungal strains for *Aspergillusniger*, *Aspergillus flavus*, and *Monascus purpureus* and NO radical scavenging antioxidant activity. Even the theoretical precisions from molecular docking study against microbial DNA gyrase corroborated well, these experimental activities provide insight into the various interactions between the ligands

and the enzyme active sites and thereby provide guidelines to design novel potent inhibitors.

**Experimental.** All the chemicals and solvents used were of synthetic grade unless otherwise noted; -9,10-difluoro-3,7-dihydro-3-methyl-7-oxo-2*H*-[1,4]oxazino[2,3,4-*ij*]quinoline-6-carboxylic acid, 2,3,4,5-tetrafluorobenzoic acid, 3,4,5,6-tetrafluorobenzene-1,2-dioic acid, 2,4-dichloro-5-fluorobenzoic acid, 2,3,4-trichloro-5-fluorobenzoic acid, pyrazine-2-carboxylic acid, 7-methyl-2-propyl-3*H*-benzo[*d*]imidazole-5-carboxylic acid, 1-(2,6-difluorobenzyl)-1*H*-1,2,3-triazole-4-carboxylic acid, PPA, *N*-methyl-*o*-phenylenediamine hydrochloride, 2-amino-phenol, and sodium bicarbonate were purchased from Sigma Aldrich, Newland Lab Ltd, Mankind Lab Ltd and Acros organics. MDC, ethyl acetate, and hexane were used from Merck and DM water from commercial source. Proton magnetic resonance <sup>1</sup>H-NMR: Bruker Avance 400 MHz NMR Spectrometer, chemical shifts are reported in ppm downfield from internal standard tetramethylsilane. IR: Thermo Scientific Nicolet iS50 FTIR Spectrometer and MS: Thermo Scientific LC–MS Orbitrap-based systems. TLC Merck TLC Silica Gel 60 F254, (0.25 mm) detection: UV light at 254 nm. Plates were visualized by UV light and iodine vapor.

### General process for the synthesis of 2-substituted benzimidazole and benzoxazole derivatives.

Polyphosphoric acid (PPA) (3 vol) was heated at 80–85°C and charged *N*-methyl-*o*-phenylenediamine hydrochloride/2-amino-phenol (1.05 mmol) and stirred for 15 min; then charged aromatic aldehyde/acid (1.0 mmol) into a reaction mass at 80–85°C. Reaction mass was maintained at 145–150°C for 3–6 h. Completion of reaction was confirmed by TLC. Twenty volume of water was added slowly to the reaction mass at 80–85°C, stirred for 1 h, and then cooled to 40°C; again, 10 vol of water

was added to the reaction mass, and pH was adjusted to 5–6 with 40% aqueous sodium hydroxide solution. Centrifuged the reaction mass and solid washed with 7 vol of water. Wet cake was heated to 70–80°C in 10 vol of water, stirred for 1 h, cooled, centrifuged, and washed with 7 vol of water, and then dried the material at 80–85°C under vacuum for 6 h. Some of these crude solid products were purified by column chromatography using hexane: ethyl acetate (8:2) as eluent to obtain pure product. These conditions are also possessing environmentally, friendly, cost-effective, and high generality.

**Spectral data.** **9,10-Difluoro-2,3-dihydro-3-methyl-6-(1-methyl-1H-benzof[d]imidazol-2-yl)-[1,4]oxazino[2,3,4-ij]quinolin-7-one (BK-1).** Yield: 83%, Color: Pale yellow solid, Molecular Formula:  $C_{20}H_{15}F_2N_3O_2$ , Mol. Wt.: 367.34 g mol<sup>-1</sup>, Elemental composition (%): *Calculated*: C, 65.39; H, 4.12; F, 10.34; N, 11.44; O, 8.71, *Found*: C, 65.50; H, 4.06; F, 10.41; N, 11.39; O, 8.64, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 8.22(s, 1H, H-Ar), 7.87–7.92 (dd, 1H, H-Ar), 7.76–7.78 (s, 1H, H-Ar), 7.43–7.45 (d, 1H, H-Ar), 7.28–7.36 (dd, 2H, H-Ar), 4.39–4.51 (dd, 3H, -CH-, -CH<sub>2</sub>), 3.85 (s, 3H, N-CH<sub>3</sub>), 1.64–1.66 (d, 3H, -CH<sub>3</sub>), <sup>13</sup>C-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 173.12, 149.99, 142.82–142.90, 136.87, 124.69, 122.21–122.71, 119.26, 112.80, 109.75, 105.40–105.59, 68.83, 54.53, 31.97, 18.12, IR  $\nu$  (cm<sup>-1</sup>): 2923 m, 1622 m, 1596 m, 1572s, 1507s, 1482s, 1349w, 1276w, 1232w, 1191w, 1110 m, 1066s, 1051s, 1007s, 898w, 839 m, 792 m, 741 s, MS (ESI):  $m/z$  368.10 (M + 1).

**2-(2,3,4,5-Tetrafluoro-6-(1-methyl-1H-benzof[d]imidazol-2-yl)phenyl)-1-methyl-1H-benzof[d]imidazole (BK-2).** Yield: 84%, Color: Off white solid, Molecular Formula:  $C_{22}H_{14}F_4N_4$ , Mol. Wt.: 410.367, Elemental composition (%): *Calculated*: C, 64.39; H, 3.44; F, 18.52; N, 13.65, *Found*: C, 64.50; H, 3.39; F, 18.49; N, 13.62, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 7.283–7.667 (m, 8H, H-Ar), 3.77–3.78 (s, 6H, N-CH<sub>3</sub>), <sup>13</sup>C-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 150.74, 142.82–142.90, 138.00, 136.24, 134.93, 122.59–122.66, 119.31, 114.52–114.74, 31.80, MS (ESI):  $m/z$  411.10 (M + 1).

**2-(2,3,4,5-Tetrafluorophenyl)-1-methyl-1H-benzof[d]imidazole (BK-3).** Yield: 92%, Color: Off white solid, Molecular Formula:  $C_{14}H_8F_4N_2$ , Mol. Wt.: 280.22, Elemental composition (%): *Calculated*: C, 60.01; H, 2.88; F, 27.12; N, 10.00, *Found*: C, 60.09; H, 2.83; F, 27.09; N, 9.99, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 7.854–7.878(s, 1H, H-Ar), 7.362–7.488 (m, 4H, H-Ar), 3.810–3.817 (s, 3H, N-CH<sub>3</sub>), <sup>13</sup>C-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 154.67, 147.99, 143.38, 142.63, 136.67, 135.44, 123.51, 122.80, 120.33–120.42, 119.29, 30.99, IR  $\nu$  (cm<sup>-1</sup>): 2919 m, 1541 m, 1484s, 1457s, 1388 m, 1328 m, 1270 m, 1206w, 1156 m, 1090 m, 1009s, 931w, 906 m, 878 s, 866 m, 768w, 742 s, 733 m, MS (ESI):  $m/z$  281.0713 (M + 1).

**2-(2,4-Dichloro-5-fluorophenyl)-1-methyl-1H-benzof[d]imidazole (BK-4).** Yield: 88%, Color: Pale yellow to off white solid, Molecular Formula:  $C_{14}H_9Cl_2FN_2$ , Mol. Wt.: 295.14, Elemental composition (%): *Calculated*: C, 56.97; H, 3.07; Cl, 24.02; F, 6.44; N, 9.49, *Found*: C, 56.91; H, 3.11; Cl, 24.04; F, 6.42; N, 9.52, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 7.85–7.87 (s, 1H, H-Ar), 7.63–7.65 (s, 1H, H-Ar), 7.352–7.469 (m, 4H, H-Ar), 3.727 (s, 3H, N-CH<sub>3</sub>), <sup>13</sup>C-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 158.00, 155.49, 149.31, 142.80, 135.68, 131.44, 130.10–129.72, 124.19–122.79, 120.42–120.18, 109.79, 30.68, IR  $\nu$  (cm<sup>-1</sup>): 3047 m, 2948w, 1599 m, 1570 m, 1485s, 1435 m, 1471s, 1383 m, 1351 m, 1326 m, 1237 m, 1129w, 1090 m, 1073 m, 1063 m, 972w, 937w, 889 s, 854 m, 799 m, 765w, 745 s, MS (ESI):  $M + m/z$  295.0323.

**2-(2,3,4-Trichloro-5-fluorophenyl)-1-methyl-1H-benzof[d]imidazole (BK-5).** Yield: 89%, Color: Light brown solid, Molecular Formula:  $C_{14}H_8Cl_3FN_2$ , Mol. Wt.: 329.58, Elemental composition (%): *Calculated*: C, 51.02; H, 2.45; Cl, 32.27; F, 5.76; N, 8.50, *Found*: C, 51.06; H, 2.36; Cl, 32.32; F, 5.78; N, 8.48, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 7.854–7.874 (s, 1H, H-Ar), 7.383–7.481 (m, 4H, H-Ar), 3.725 (s, 3H, N-CH<sub>3</sub>), IR  $\nu$  (cm<sup>-1</sup>): 3078 m, 2921 m, 1786w, 1517w, 1470 m, 1439s, 1435 m, 1415s, 1400s, 1358 m, 1330 m, 1233 m, 1194 m, 1093 m, 1006 m, 965 m, 880 m, 801 m, 781 m, 746 s, 732 s, MS (ESI):  $m/z$  331.1588 M + 1.

**1-Methyl-2-(pyrazin-2-yl)-1H-benzof[d]imidazole (BK-6).** Yield: 95%, Color: Pale yellow solid, Molecular Formula:  $C_{12}H_{10}N_4$ , Mol. Wt.: 210.2346, Elemental composition (%): *Calculated*: C, 68.56; H, 4.79; N, 26.65, *Found*: C, 68.61; H, 4.77; N, 26.62, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 9.666–9.669(s, 1H, H-Ar), 8.626–8.664 (dd, 2H, H-Ar), 7.868–7.889 (s, 1H, H-Ar), 7.458–7.479 (s, 1H, H-Ar), 7.338–7.416 (m, 2H, H-Ar), 4.273 (s, 3H, N-CH<sub>3</sub>), <sup>13</sup>C-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 147.57, 146.37–146.06, 144.00, 142.72–142.60, 137.24, 123.98, 123.04, 120.38, 109.99, 32.67, IR  $\nu$  (cm<sup>-1</sup>): 3506 m, 3320w, 3060w, 1613w, 1531 m, 1473 m, 1440 m, 1402 m, 1367 m, 1329 m, 1283w, 1261w, 1218w, 1187 m, 1048, 1081 m, 1038s, 1017 m, 927w, 900w, 845 m, 828 m, 765 m, 757 m, 737 s, 723 m, MS (ESI):  $m/z$  211.2335 M + 1.

**6-(Benzof[d]oxazol-2-yl)-4-methyl-2-propyl-1H-benzof[d]imidazole (BK-7).** Yield: 85%, Color: Off white solid, Molecular Formula:  $C_{18}H_{17}N_3O$ , Mol. Wt.: 291.3471, Elemental composition (%): *Calculated*: C, 74.20; H, 5.88; N, 14.42; O, 5.49, *Found*: C, 74.12; H, 5.94; N, 14.40; O, 5.54, <sup>1</sup>H-NMR (400 MHz, CDCl<sub>3</sub>, ppm):  $\delta$  = 8.278 (s, 1H, H-Ar), 7.996 (s, 1H, H-Ar), 7.758–7.735 (s, 1H, H-Ar), 7.598–7.576 (s, 1H, H-Ar), 7.371–7.286 (m, 2H, H-Ar), 4.745–4.726 (s, 1H, N-H), 2.954–2.916 (t, 2H, -CH<sub>2</sub>), 2.669–2.633 (s, 3H, CH<sub>3</sub>-Ar), 1.921–1.847 (t, 2H, -CH<sub>2</sub>), 1.031–0.996 (s, 3H, -CH<sub>3</sub>),

$^{13}\text{C}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 164.31, 157.20, 150.68, 141.99, 124.74–124.51, 122.38, 120.80, 119.31, 110.52, 31.30, 21.77, 17.14, 13.85, IR  $\nu$  ( $\text{cm}^{-1}$ ): 3131 m, 2966 m, 1616w, 1556 m, 1455s, 1379w, 1325 m, 1248s, 1176s, 1005 m, 866 m, 842 m, MS (ESI):  $m/z$  390.1823.

**2-(1-(2, 6-Difluorobenzyl)-1H-1,2,3-triazol-4-yl)benzo[d]oxazole (BK-8).** Yield: 88%, Color: White solid, Molecular Formula:  $\text{C}_{16}\text{H}_{10}\text{F}_2\text{N}_4\text{O}$ , Mol. Wt.: 312.2736, Elemental composition (%): Calculated, 61.54; H, 3.23; F, 12.17; N, 17.94; O, 5.12, Found: C, 61.60; H, 3.21; F, 12.20; N, 17.90; O, 5.09,  $^1\text{H}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 8.298–8.334 (s, 1H, H-Triazole ring), 7.779–7.748 (s, 1H, H-Ar), 7.659–7.617 (s, 1H, H-Ar), 7.472–7.286 (m, 3H, H-Ar), 7.059–6.896 (s, 2H, H-Ar), 5.781 (s, 2H,  $-\text{CH}_2-$ ),  $^{13}\text{C}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 162.66–162.59, 160.15, 155.89, 150.33, 141.34, 137.53, 132.07–131.86, 125.60, 124.90–124.74, 120.03, 112.14–111.89, 111.01, 110.93, 110.26–110.07, 41.88–41.80, IR  $\nu$  ( $\text{cm}^{-1}$ ): 3068 m, 1630s, 1593 m, 1508w, 1471s, 1452 m, 1437 m, 1379w, 1272w, 1234 m, 1177w, 1141 m, 1044 m, 1027s, 1005 m, 990 m, 919w, 891 m, 841w, 791 s, 757 s, 744 s, MS (ESI):  $m/z$  313.3 M + 1.

**2-(2-(Benzo[d]oxazol-2-yl)-3,4,5,6-tetrafluorophenyl)benzo[d]oxazole (BK-9).** Yield: 85%, Color: White solid, Molecular Formula:  $\text{C}_{20}\text{H}_8\text{F}_4\text{N}_2\text{O}_2$ , Mol. Wt.: 384.28, Elemental composition (%): Calculated: C, 62.51; H, 2.10; F, 19.78; N, 7.29; O, 8.33, Found: C, 62.56; H, 2.08; F, 19.81; N, 7.25; O, 8.30,  $^1\text{H}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 7.727–7.702 (d, 2H, H-Ar), 7.445–7.283 (m, 6H, H-Ar),  $^{13}\text{C}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 154.32, 150.81, 141.19, 126.24, 124.93, 120.87, 110.83, IR  $\nu$  ( $\text{cm}^{-1}$ ): 2923 m, 1605w, 1562 m, 1507 m, 1478s, 1450 m, 1393w, 1238 m, 1170 m, 1031w, 1004 m, 944 m, 846 s, 799w, 768 m, 738 s, MS (ESI):  $m/z$  385.05 M + 1.

**2-(2,4-Dichloro-5-fluorophenyl)benzo[d]oxazole (BK-10).** Yield: 89%, Color: Light brown solid, Molecular Formula:  $\text{C}_{13}\text{H}_6\text{Cl}_2\text{FNO}$ , Mol. Wt.: 282.09, Elemental composition (%): Calculated: C, 55.35; H, 2.14; Cl, 25.14; F, 6.73; N, 4.97; O, 5.67, Found: C, 55.40; H, 2.10; Cl, 25.18; F, 6.75; N, 4.93; O, 5.64,  $^1\text{H}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 8.055–8.031 (d, 1H, H-Ar), 7.892–7.869 (d, 1H, H-Ar), 7.679–7.642 (t, 2H, H-Ar), 7.479–7.431 (t, 2H, H-Ar), IR  $\nu$  ( $\text{cm}^{-1}$ ): 3250w, 1691w, 1637w, 1342w, 1270 m, 1235 m, 1180w, 1101s, 1089s, 1069s, 1022 m, 994 m, 912 m, 890 m, 818w, 790 m, 759 s, MS (ESI):  $m/z$  283.70 M + 1.

**2-(2,3,4-Trichloro-5-fluorophenyl)benzo[d]oxazole (BK-11).** Yield: 94%, Color: Pale yellow to off white solid, Molecular Formula:  $\text{C}_{13}\text{H}_5\text{Cl}_3\text{FNO}$ , Mol. Wt.: 316.54, Elemental composition (%): Calculated: C, 49.33; H, 1.59; Cl, 33.60; F, 6.00; N, 4.42; O, 5.05, Found: C, 49.38; H, 1.53; Cl, 33.64; F, 5.94; N, 4.44; O, 5.07,  $^1\text{H}$ -NMR (400 MHz,  $\text{CDCl}_3$ , ppm):  $\delta$  = 7.988–7.966 (s, 1H, H-Ar), 7.891–7.889 (d, 1H, H-Ar), 7.667–7.646 (d, 1H,

H-Ar), 7.495–7.422 (t, 2H, H-Ar), IR  $\nu$  ( $\text{cm}^{-1}$ ): 3084 m, 2971w, 1778w, 1588w, 1524 m, 1473w, 1450 m, 1436s, 1419s, 1399 m, 1359 m, 1341 m, 1307w, 1245 m, 1233w, 1190w, 1133 m, 1100 m, 1069w, 1002w, 933w, 918 m, 889 m, 876 m, 846 m, 785 m, 761 m, 744 s, MS (ESI):  $m/z$  317.40 M + 1.

#### Experimental protocol for biological activity.

**Antimicrobial activity.** The bacterial strains *Escherichia coli* MTCC 1696, *Bacillus subtilis* MTCC 2616, *Staphylococcus aureus* MTCC 3160, and *Pseudomonas aeruginosa* MTCC 424 and fungal strains *Aspergillus niger* MTCC 281, *Aspergillus flavus* MTCC 277, and *Monascus purpureus* MTCC 369 were obtained from microbial type culture collection, Institute of Microbial Technology, Chandigarh India. The bacterial strains were cultured in nutrient agar medium at 37°C for 24 h, and the fungal strains were cultured in potato dextrose medium for 3 days at 28°C. The culture slants were maintained for 1 month at 4°C and subcultured after each 1 month. All the media were obtained from Titan Biotech, New Delhi, for microbiological culture purpose. Ciprofloxacin 1 mg/mL, miconazole 1 mg/mL, and fluconazole 1 mg/mL (Himedia, Mumbai) solutions in dimethyl sulphoxide (DMSO) (SD Fine Chem., Mumbai, India) were used as standard antibacterial and antifungal drugs, respectively. DMSO was used as control. The newly synthesized compounds were tested against a panel of bacterial strains (*Escherichia coli* MTCC 1696, *Bacillus subtilis* MTCC 2616, *Staphylococcus aureus* MTCC 3160, and *Pseudomonas aeruginosa* MTCC 424) and fungal strains (*Aspergillus niger* MTCC 281, *Aspergillus flavus* MTCC 277 and *Monascus purpureus* MTCC 369) [30,31] by cup-plate method. Broth inoculums of different bacteria were standardized to 0.5 McFarland standards by means of dilution with sterile nutrient broth, measuring the optical density measurement at 625-nm wavelength. Absorbance readings were fixed to be within the range of 0.08–0.13 (equivalent to approximately  $1.5 \times 10^8$  CFU/mL). Broth inoculums of different fungal strains were prepared in potato dextrose broth to CFU of  $2.0 \times 10^4$ /mL. Potato dextrose agar and nutrient agar were used as culture medium for antifungal and antibacterial activities, respectively. Using an agar punch, wells (0.6-cm diameter) were made on seeded agar plates, and a concentration of 0.5, 20, and 50  $\mu\text{g/mL}$  of test compounds in DMSO (100  $\mu\text{L}$ ) was added into each well, labeled previously. A control was also prepared using solvent DMSO. The petri-discs were incubated at 28°C for 72 h for fungi and 37°C for 24 h for bacteria. Antimicrobial activity was determined by measuring zone of inhibition in millimeters, and data have been presented as ( $n = 3$ , mean) and compared with standard fluconazole and miconazole (1  $\mu\text{g/mL}$ ) and ciprofloxacin (1  $\mu\text{g/mL}$ ). Test was performed in quadruplets. Synthesized chemicals that showed antimicrobial

activities against the bacterial and fungal strains tested were subjected to broth micro dilution assay to evaluate their respective MIC value.

**Antioxidant screening. DPPH radical scavenging activity.** The molecule 1, 1-diphenyl-2-picrylhydrazyl (a,a-diphenyl-bpicrylhydrazyl; DPPH) is characterized as a stable free radical by virtue of the delocalization of the spare electron over the molecule as a whole, so that the molecule does not dimerize, as would be the case with most other free radicals [32,33]. The delocalization of electron also gives rise to the deep violet color, characterized by an absorption band in ethanol solution centered at about 517 nm. When a solution of DPPH is mixed with that of a substrate (AH) that can donate a hydrogen atom, then this gives rise to the reduced form with the loss of this violet color. Briefly, 1 mL of synthesized compounds as 1 mM was mixed with 3.0 mL of DPPH (0.5 mmol/L in methanol); the resultant absorbance was recorded at 517 nm after 30 min of incubation at 37°C. The percentage of scavenging activity was derived using the following formula:

$$\text{Percentage of inhibition (\%)} = \frac{[(A \text{ control} - A \text{ sample}) / A \text{ control}] \times 100}{}$$

where A control is the absorbance of DPPH and A sample is the absorbance reaction mixture (DPPH with sample).

**Nitric oxide radical scavenging activity.** Nitric oxide (NO) is generated in biological tissues by specific NO synthases, which metabolizes arginine to citrulline with the formation of NO via a five electron oxidative reaction. The compound sodium nitroprusside is known to decompose in aqueous solution at physiological pH (7.2) producing NO under aerobic conditions; NO reacts with oxygen to produce stable products (nitrate and nitrite), the quantities of which can be determined using Griess reagent.

One millimeter of 10 mmol of sodium nitroprusside dissolved in 0.5 mL of phosphate buffer saline (pH 7.4) was mixed with 1 mL of 1 mmol synthetic compound in DMSO. The mixture was incubated at 25°C for 150 min. After incubation, the reaction mixture was mixed with 1.0 mL of pre-prepared Griess reagent [(1.0 mL sulfanilic acid reagent) 0.33% in 20% glacial acetic acid at room temperature for 5 min with 1 mL of naphthylethylenediamine dichloride (0.1% w/v)]. The mixture is then incubated at room temperature for 30 min, and its absorbance pouring into a cuvette is measured at 546 nm. The decreasing absorbance indicates a high NO scavenging activity.

The amount of NO radical inhibition is calculated by the following equation:

$$\% \text{inhibition of NO radical} = (A_0 - A_1) / A_0 \times 100 \quad (1)$$

where  $A_0$  is the absorbance before reaction and  $A_1$  is the absorbance after reaction has taken place with Griess reagent.

**Computational study. Molecular docking.** The Glide (Grid-Based Ligand Docking with Energetics) module of the Schrödinger molecular modeling package (Schrödinger, LLC, New York, NY) [34] was used to investigate the binding mode of the benzimidazole and benzoxazole derivatives discussed herein into the active site of microbial DNA gyrase enzyme. For this, the 3D-crystal structure of *DNA gyrase subunit b* complexed with an inhibitor was retrieved from the protein data bank (pdb code: 1KZN) and refined using the Protein Preparation Wizard. This preprocessing includes eliminating the crystallographically observed water molecules (water without H bonds) as they were not found to be conserved in the ligand interaction, assignment of the correct bond orders, creating disulfide bonds, identifying overlaps, and adding hydrogen atoms corresponding to pH 7.0 considering the appropriate ionization states for the basic and acidic amino acid residues. After assignment of appropriate charge and protonation state, finally, the prepared enzyme structure was subjected to energy minimization to relieve the steric clashes among the residues due to addition of hydrogen atoms, using OPLS-2005 force field until the average Root Mean Square Deviation (RMSD) of the non-hydrogen atoms reached 0.3 Å. On the other hand, the 2D structures of the benzimidazole and benzoxazole derivatives were sketched with the build panel and converted to energy minimized 3D structures using ligand preparation panel in Maestro. It also involves addition of hydrogens, correction of chiralities and ionization states, generation of tautomers, and ring conformations, adjusting the realistic bond lengths and bond angles. The partial charges were ascribed to these structure using the OPLS-2005 force-field followed by energy minimization until their average RMSD reached 0.001 Å. The shape and properties of the active site of DNA gyrase enzyme were characterized and set up for docking using the receptor grid generation panel which uses the native ligand as the reference coordinate because it signifies the binding sites of a molecule with respect to the target. For this, a grid box of  $12 \times 12 \times 12$  Å around the centroid of the native ligand was generated. These energy minimized enzyme and ligands structures were then used as input for the molecular docking against the defined active site of DNA gyrase enzyme using with extra precision (i.e., GlideXP) scoring function to gauge the enzyme-ligand binding affinities and mode of interaction. The output file (docking poses) generated for these ligands was visualized through the Maestro's Pose Viewer utility for the mode of binding and analyzed for the key thermodynamic interactions (bonded and nonbonded) with the active site residues.

**Absorption, distribution, metabolism, and excretion properties.** The success of a drug is determined not only

by good efficacy but also by an acceptable ADME profile. In this study, molecular volume (MV), molecular weight (MW), logarithm of partition coefficient (miLog *P*), number of hydrogen bond acceptors (n-ON), number of hydrogen bonds donors (n-OHNN), topological polar surface area (TPSA), number of rotatable bonds (n-ROTB), and Lipinski's rule of five [30] using Molinspiration online property calculation toolkit [35a] was calculated. Absorption (% ABS) was calculated by: %ABS = 109-(0.345 × TPSA) [35b].

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